

PHANENDRA PALADUGU

Griffith, Indiana | (219) 455-0679 | phanendraus111@gmail.com | linkedin.com/in/phanendra-p | github.com/phanipal

PROFESSIONAL SUMMARY

GenAI / LLM Engineer with 5+ years of experience building NLP, document intelligence, and retrieval-based AI systems using Python, Hugging Face, LangChain, vector databases, and open-source LLMs. Hands-on with RAG pipelines, embeddings, NER, document classification, LLM fine-tuning (QLoRA/GRPO), and API deployment using FastAPI, Docker, Kubernetes, and CI/CD. Strong at turning unstructured enterprise documents into searchable, reliable AI workflows with measurable retrieval and model evaluation.

TECHNICAL SKILLS

Languages: Python, SQL, Bash

GenAI & LLM: LangChain, LangGraph, AI Agents, Function Calling, Tool Use, Llama 3.1, Hugging Face Transformers, Sentence Transformers, RAG, Prompt Engineering, Few-Shot Learning

LLM Fine-Tuning & Local Deployment: QLoRA, GRPO, GGUF Quantization, Qwen2.5-Coder, Ollama, llama.cpp, HuggingFace Hub

NLP & Search: NLTK, SpaCy, NER, FAISS, ChromaDB, Embeddings, Cosine Similarity, TF-IDF, Text Classification

ML & DL: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost

Data Engineering: Pandas, NumPy, PySpark, Apache Spark, ETL Pipelines, MySQL, Teradata, SAP BW/APO

MLOps & Deployment: FastAPI, Flask, Docker, Kubernetes, Git, CI/CD, LLMOps, MLflow, Evidently AI, JWT Auth, pytest, GitHub Actions

Cloud: AWS (S3, SageMaker, EC2), Azure (Azure OpenAI, Azure AI Search), GCP (BigQuery, Vertex AI)

Collaboration: Agile/Scrum, JIRA, Confluence, Cross-Functional Partnership, Technical Documentation

EXPERIENCE

GenAI Engineer | Wells Fargo

Jan 2025 - Present

- Built RAG-based document Q&A workflows using LangChain, ChromaDB, and Sentence-Transformers to help teams search internal policy documents, compliance manuals, and regulatory filings in natural language.
- Developed NLP pipelines for document classification, entity extraction, and report summarization using SpaCy, Hugging Face Transformers, and fine-tuned language models.
- Deployed LLM applications as JWT-secured FastAPI services using Docker and Kubernetes, adding response validation, citation checks, and basic guardrails for compliance-focused use cases.
- Tested paragraph-level, sliding-window, and semantic chunking strategies with Sentence-Transformers, ChromaDB, and FAISS to improve retrieval quality across different document types.
- Built LangChain and LangGraph agent workflows for compliance research and document analysis that chain retrieval, function calling, and API/database lookups, with use-case-specific prompts and guardrails.
- Created evaluation checks for retrieval quality, answer accuracy, and latency using metrics such as Recall@K and MRR, and worked with compliance and legal teams to review content filtering and response validation.

Lab Assistant | Purdue University

Aug 2024 - Dec 2024

- Coached students through electrical measurements (voltage, current, impedance, frequency response) using oscilloscopes, multimeters, and signal generators across varied lab scenarios.
- Guided circuit design, wiring, and troubleshooting, applying systematic debugging to isolate and resolve hardware faults.
- Reviewed lab notebooks and technical reports, giving detailed feedback on data accuracy, documentation, and analytical reasoning.

Technology Analyst | Infosys

Jan 2020 - Dec 2021

- Built a TensorFlow/Keras NLP pipeline to classify 200+ monthly ServiceNow incidents, extract intent, and recommend resolution articles using text embeddings and cosine similarity, reducing manual triage by 30%.
- Created a knowledge-article retrieval system with TF-IDF and semantic text matching to surface relevant documentation for support tickets, an early precursor to modern RAG workflows.
- Implemented text-preprocessing and feature-extraction pipelines with NLTK, SpaCy, and Pandas for tokenization, named entity recognition, and embedding generation on unstructured enterprise text.
- Integrated NLP models into Flask REST APIs for real-time inference, with response logging, versioned endpoints, and reusable request templates for internal support workflows.
- Built a real-time email-monitoring application that used rule-based NLP to classify support queries, identify intent, and create ServiceNow tickets for automated processing.
- Partnered with planning teams to validate data consistency, support KPI tracking, and troubleshoot data anomalies across global operations.

Senior Systems Engineer | Infosys

Jan 2017 - Dec 2019

- Automated recurring SAP BW/APO data and ETL tasks using Python and Selenium, reducing manual effort by 800+ hours annually.
- Created a Python email-monitoring application with rule-based NLP and text pattern matching to auto-classify queries, extract intent, and generate ServiceNow tickets for 24/7 automated processing.
- Established structured data pipelines using SAP BW InfoCubes and DataSources to ensure clean feeds for reporting and downstream model consumption.
- Wrote complex SQL for extraction and validation across SAP BW, supporting ad-hoc reporting for global supply-chain operations.

PROJECTS

Slonik-7B-GRPO: Text-to-SQL Fine-Tuned LLM

Python, PyTorch, QLoRA, GRPO, Qwen2.5-Coder-7B, BIRD-SQL, PostgreSQL, SQLite, GGUF, Ollama, HuggingFace

- Fine-tuned Qwen2.5-Coder-7B using QLoRA and GRPO for PostgreSQL/SQLite text-to-SQL. Improved BIRD-PG accuracy from 12.22% to 38.20%, outperforming GPT-4o (34.44%) at 7B parameters, and published GGUF quantizations for local deployment via Ollama and llama.cpp. HuggingFace: huggingface.co/Phani-labs/Slonik-7B-GRPO

DocBrain: Enterprise Document Intelligence Engine

Python, PyTorch (CUDA), ChromaDB, Sentence-Transformers, Llama 3.1 8B, Tesseract/DocTR, SpaCy, FastAPI, Streamlit, Docker

- Built a RAG-based document intelligence pipeline across four document types using ChromaDB, Sentence-Transformers, and Llama 3.1 8B. Evaluated on 30 hand-written queries (0.90 Recall@5, 0.78 MRR, 1.00 Recall@1 on contracts), with OCR routing, SpaCy NER, faithfulness checks, and 162 pytest tests.

IntelliDesk: AI-Powered IT Service Desk

Python, FastAPI, XGBoost, TF-IDF, Sentence-Transformers, FAISS, MLflow, Streamlit, Evidently AI, Docker, GitHub Actions

- Built an ML ticket-triage system using XGBoost classifiers for category and priority. Addressed a 12:1 class imbalance with sqrt-scaled class weights, added a Sentence-Transformer + FAISS retriever for similar-ticket lookup, and tracked experiments in MLflow with Evidently drift monitoring.

EDUCATION

Master of Science, Electrical and Computer Engineering

GPA: 3.61/4.0

Purdue University Northwest, Hammond, Indiana

2025

Coursework: Machine Learning, Neural Networks, Big Data, Computer Vision, Digital Signal Processing